



STATEMENT OF WORK

U.S. DEPARTMENT OF STATE

OBO FACILITIES MANAGEMENT DIVISION

Fireproofing walls and installation of Fire-Rated doors

Madrid Embassy Sub-Basement

U.S. Embassy Madrid

Spain

1. INTRODUCTION / SUMMARY OF WORK PROJECT

1.1. Purpose

The purpose of this Statement of Work (SOW) is to define the requirements for the design, supply, and installation of 9 units of two-hour (2 HR) UL-classified fire-rated assemblies around critical electrical and mechanical equipment located in the sub-basement of the U.S. Embassy in Madrid, Spain. The work shall enhance fire compartmentation, life safety, and compliance with applicable U.S. and Spanish fire codes.

Address: The site is located at Serrano 75. Madrid, Spain

1.2. Scope

The Contractor shall furnish all labor, materials, equipment, and supervision to install fire-rated walls, doors, frames, and firestopping in the designated areas, achieving a continuous two-hour fire-resistance rating from finished floor to the underside of the concrete deck above.

This project includes:

- Supply and installation of three (3) double-leaf and six (6) single-leaf 2 HR fire-rated door assemblies.
- Fire-rated wall construction around the 15 kV step-down transformer room, low voltage/ATS room, oil boiler room, UPS room, and diesel fire pump room.
- Firestopping of all penetrations with UL-approved systems. Please note this may include rebuilding large sections of walls.

2. PRIOR TO IMPLEMENTATION

- Required site visit and survey. The contractor shall be responsible for researching the existing conditions and matters that affect the cost or performance of the services. Contractor is required to attend a site visit to confirm all measurements and survey existing layout prior to submission of BOM and quotation. Measurements provided are approximate and US Embassy shall not be held responsible for inaccurate dimensions.
- The contractor shall coordinate and obtain any permit required by local authorities for the execution of the project.
- DOCUMENT SUBMITTAL PACKAGE. Submit before awarding process, document submittal package that includes:
 - .1. Product Data: Manufacturer's data sheets on each product to be used, including:
 - .2. Shop Drawings: Show typical details of assembly, erection, and anchorage.
 - .3. Complete finalized Work Plan including Critical Path Method (CPM) schedule.

- DOCUMENT SUBMITTAL PACKAGE AFTER AWARD. Submit to the CO and/or COR approval prior to installation:
 - .1. Certificates of Compliance – UL listings and fire ratings.
 - .2. As-Built Drawings – Final installed condition showing rated assemblies.
 - .3. Inspection Reports – From certified inspectors.

3. DESCRIPTION OF THE WORK

3.1. Area isolation and access restriction

- The contractor shall provide a plan for the isolation and restriction of access to the area to be intervened.
- Works must only be carried out by staff with technical knowledge to do so and always complying with the security rules.
- The contractor shall take all necessary precautions to protect existing facilities, including but not limited to equipment, pipes, fire devices, furniture, etc. During the removal of existing frames and doors, the contractor must exercise utmost care to minimize damage to the surrounding areas.
- The contractor shall remove and relocate furniture as required to protect from damages during work at each area.
- No construction work shall begin until approvals of the Pre-Construction Submittals are accepted by the COR.

3.2. Fire-Rated Wall Assemblies

- Construct new UL-classified 2 HR fire-rated partitions from finished floor slab to the underside of the structural concrete deck above.
- The Contractor shall construct a new fire-rated wall to allow installation of the new fire-rated doors N4 and frames. All wall assemblies shall be built to meet the same fire rating requirements as the doors and installed to ensure the full integrity and compliance of the fire barrier.
- Additionally, the Contractor shall construct fire-rated wall sections around the other fire-rated doors and frames, once replaced, to close the existing oversized openings. All wall assemblies shall be built to meet the same fire rating requirements as the doors and installed to ensure the full integrity and compliance of the fire barrier.
- Acceptable construction types include CMU block walls or metal stud framing with fire-rated gypsum board systems per UL design details.
- All seams, joints, and edges shall be sealed with fire-rated sealants to maintain the required rating.

- Part of the false ceiling shall be demolished and the entire interior partition wall of the locker room shall be lined with EI-120 material up to the slab.
- The interior door in the UPS room shall be closed off with an EI 120 partition wall.

3.3. Fire Door Assemblies

- Supply fire-rated doors, ensuring compliance with local, national, and international safety standards. All doors must comply with NFPA 80 and local EN standard. Follow the relevant fire resistance standards in EN 1634-1 for the doors and frames.
- Supply three (3) double-leaf and six (6) single-leaf fire-rated hollow metal door assemblies, factory-fabricated and UL 2 HR labeled.
- Standard measurements are allowed.
 - .1. Double-leaf:
 - Door N1. Boiler room West. Dimension 1680x2100 mm
 - Door N5. High Voltage room. Dimension 1680x2100 mm
 - Door N8. Mechanical room entrance door. Dimension 1680x2100mm
 - .2. Single leaf:
 - Door N2. Boiler room to locker room. Opening dimension 1380x2000mm. Supply standard door 900x2000mm. Wall is required to install a door with standard width.
 - Door N3. Boiler room East. Dimension 930x2100mm
 - New door N4 Fire Pump. Opening dimension: 1400x2430mm. Supply standard door 900x2000mm. Construction of a wall is required to install a door with standard width.
 - Door N7a and 7b. Low Voltage Electrical room. Dimension 930x2100mm (two doors).
 - Door N6. UPS. Dimension 930x2100 mm.
- Finish: Oven-baked lacquered paint gray
- Frames shall be welded or knock-down type, UL-listed, with appropriate anchors for the substrate.
- Door hardware shall include self-closing devices, positive-latching mechanisms, and panic or lever hardware as required for egress per NFPA 101 and CTE DB-SI.
- Doors are required to swing out in the direction of exit travel.

- All hallway doors must have fire exit hardware (panic bar hardware) and are required to swing out in the direction of exit travel. Doors shall be operable without keys, tools, or special knowledge.
- The doors 5, 6, 7a and 7b must be equipped with a door closer. Fire-rated polyamide handles (on exterior side of the door).
- Coordinate swing direction for unobstructed access and compliance with egress requirements.
- Sealing: complete perimeter sealing of frame-to-wall joints to ensure sectorization and compliance with explosion-resistance recommendations. Install smoke seals around the door perimeter to prevent the passage of smoke, which can be more hazardous than flames.

3.4. Firestopping

- All penetrations through fire-rated walls shall be sealed with UL-classified firestop systems tested per UL 1479 / ASTM E814, maintaining the full 2 HR rating.
- Firestop systems shall be selected based on the penetrating item type (conduit, cable tray, piping, ductwork, etc.) and substrate.

3.5. Finishes and paints

- The contractor shall do patching and painting works as needed.

4. MATERIAL REQUIREMENTS

4.1. Doors and Frames

- Hollow metal construction, minimum 1.5 mm (16 gauge) steel skins.
- UL 2 HR fire rating, labeled per NFPA 80.
- Factory primer finishes suitable for field painting.

4.2. Wall Assemblies

- CMU or UL-classified gypsum assemblies per UL Design Nos. U905, U419, or equivalent.
- Fire-resistant mineral wool insulation where required by UL design.

4.3. Firestopping

- UL-approved intumescent sealants, mortars, or putty pads as appropriate for the application.
- Firestop collars for plastic pipe penetrations.

5. EXECUTION

5.1. Disassembly

- Includes disassembly of existing doors, solidification of frame interiors, floor trims as needed (cement), masonry finishing, and adaptation to embedded frames.

5.2. Installation

- All fire doors must be installed by qualified professionals experienced with fire door installations, ensuring compliance with fire safety regulations.
- All work shall be performed in accordance with the manufacturer's instructions, applicable UL design details, and referenced codes.
- Maintain tolerances and clearances as per NFPA 80 Table 6.3.1.7.
- Coordinate with electrical, mechanical, and security trades to ensure fire-rated construction continuity.
- Fire Door Assembly: Ensure the door assemblies have frames, hardware, and seals meeting fire resistance standards.

5.3. Labeling and Certification

- Fire Door Labeling: Each installed fire door must have a visible label indicating the fire resistance rating and certification for the relevant testing standards (EN 1634-1).
- All installation work must have documented certification from manufacturers and fire resistance testing labs.
- The installation must meet the Código Técnico de la Edificación (CTE), ensuring fire resistance standards are met.

5.4. Dispose and Clean Intervened Area(s)

- Handling and dispose all debris; daily.
- The contractor will rectify any damage to any areas affected on completion of the works.
- Contractor shall do a final cleaning of the work, releasing mortars adhered to floors, sweeping and removal of debris at the foot of the load, load, and transport to the landfill.
- Clean all intervened area(s).
- All this item must include collection and treatment of all the materials from dust on container. Management and transport of inert waste, with container, to specific landfill.

6. AFTER IMPLEMENTATION

- When all site work is complete, the Contractor's site superintendent shall review all activities and walk the project area with the COR to ensure that the embassy staff is satisfied, and all work items are acceptable. If any items are not completed, they shall be noted and provided to the Contractor to be addressed and completed immediately.

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- The Contractor shall ensure that, upon demobilization, the site is left in a clean and orderly appearance, equal to or better than prior to start of work.
- The contractor shall allow for testing and commissioning of the entire system.
- Perform inspection by a Certified Fire Door Inspector and Firestop Special Inspector in accordance with NFPA 80 and ASTM E2174/E2393.
- Submit inspection reports to the Contracting Officer's Representative (COR).

7. PERIOD OF PERFORMANCE

- The project shall start within five (5) calendar days after the date the Contractor receives the Notice-to-Proceed (NTP) issued by the CO. The contractor shall sustain a Kick-off meeting to review the SOW, detailed schedule, execution, and safety plan.
- Timetable of Deliverables
 - .1. The Contractor shall provide the final proposal design of the Fire rated doors 10 days after receiving the Notice to Proceed.
 - .2. Complete the entire work no later than NINETY (90) calendar days after Notice to Proceed. The time stated for completion shall include final cleanup of the premises.

8. WARRANTY

- Shall meet Spanish construction standards

9. SAFETY PROGRAM (Abridged)

- Contractor must submit 10 days before start of the works, an Activity Hazard Analysis (AHA).
- The Contractor must review the work and write a summary of any hazards expected (AHA).
- The AHA must address what the Contractor will do to mitigate those hazards.
- This may be task items such as "Use lock-out, tag-out for electrical circuits", "Use Personal Protective Equipment (PPE) when pouring concrete", etc.
- General. The contractor shall provide and maintain work environments and procedures which will safeguard the public and Government personnel, property, materials, supplies, and equipment exposed to contractor operations and activities; avoid interruptions of Government operations and delays in project completion dates; and control costs in the performance of this contract. For these purposes, the contractor shall:
- Provide appropriate safety barricades, signs and signal lights.

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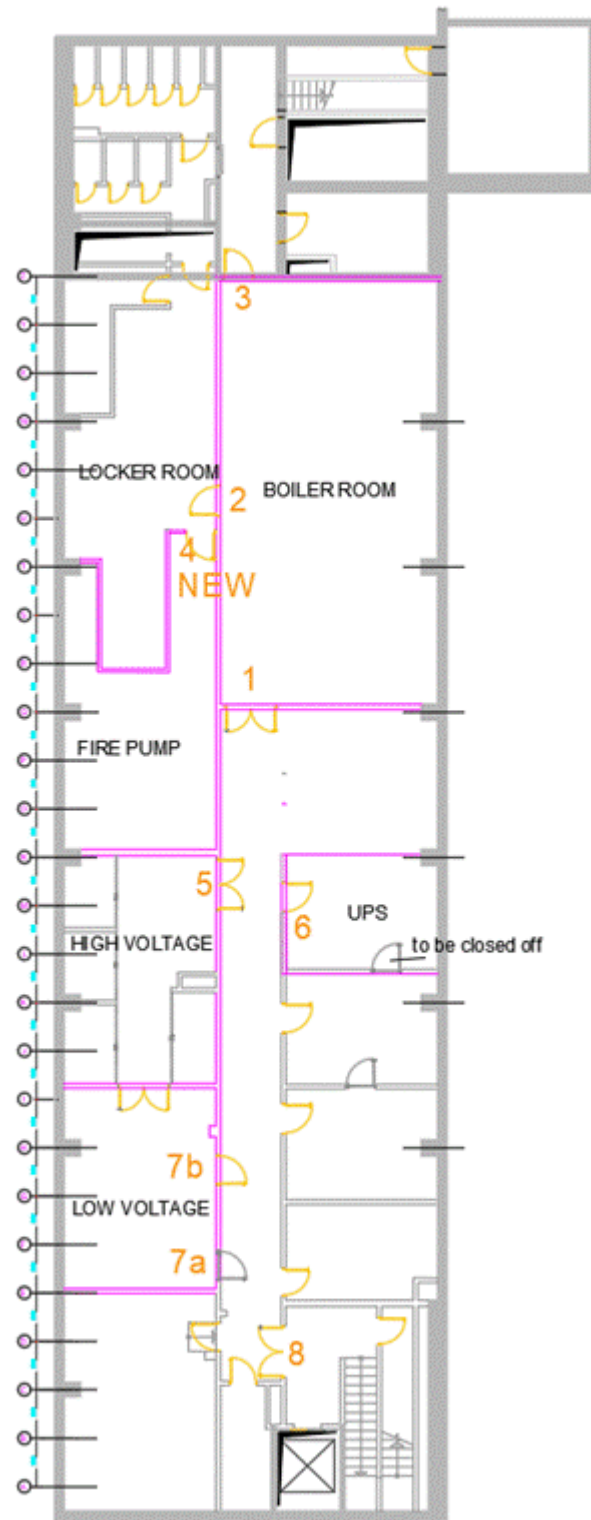
- Comply with the standards issued by US government over occupational health and safety issues; and,
- Ensure that any additional measures the contracting officer determines to be reasonably necessary for this purpose are taken.
- Additional requirements regarding safety are applicable and referenced from the Corp of Engineers Safety Manual EM 385-1-1, available on the internet, if the work involves and required:
 - .1. Scaffolding;
 - .2. Work at heights above two (2) meters;
 - .3. Trenching or other excavation greater than one (1) meter in depth;
 - .4. Earth moving equipment;
 - .5. Temporary wiring, use of portable electric tools, or other recognized electrical hazards. Temporary wiring and portable electric tools require the use of a ground fault circuit interrupter (GFCI) in the affected circuits; other electrical hazards may also require the use of a GFCI;
 - .6. Work in confined spaces (limited exits, potential for oxygen less than 19.5 percent or combustible atmosphere, potential for solid or liquid engulfment, or other hazards considered to be immediately dangerous to life or health such as water tanks, transformer vaults, sewers, cisterns, etc.);
 - .7. Hazardous materials—a material with a physical or health hazard including but not limited to, flammable, explosive, corrosive, toxic, reactive or unstable, or any operations which creates any kind of contamination inside an occupied building such as dust from demolition activities, paints, solvents, etc.; or
 - .8. Hazardous noise levels.
- Records. The contractor shall maintain an accurate record of exposure data on all accidents incident to work performed under this contract resulting in death, traumatic injury, occupational disease, or damage to or theft of property, materials, supplies, or equipment. The contractor shall report this data in the manner prescribed by the contracting officer.
- Subcontracts. The contractor shall be responsible for its subcontractors' compliance with this clause.
- Written safety program. Before commencing work, the contractor shall:
 - .1. Submit a written plan to the contracting officer for implementing this clause. The plan shall include specific management or technical procedures for effectively controlling hazards associated with the project; and,
 - .2. Meet with the contracting officer to discuss and develop a mutual understanding relative to administration of the overall safety program.
 - .3. Procedure in case of injury or emergency events.
- Notification. The contracting officer shall notify the contractor of any non-compliance with these requirements and the corrective actions required. This notice, when delivered to the contractor or the contractor's representative on site, shall be deemed sufficient notice of the non-compliance and corrective action required. After receiving the notice, the contractor

shall immediately take corrective action. If the contractor fails or refuses to promptly take corrective action, the contracting officer may issue an order suspending all or part of the work until satisfactory corrective action has been taken. The contractor shall not be entitled to any equitable adjustment of the contract price or extension of the performance schedule on any suspension of work order issued under this clause.

- The Contractor shall:
 - .1. Comply with U.S. Embassy Madrid security clearance and access requirements.
 - .2. Follow all site safety regulations, including dust control, PPE use and hot work permitting.
 - .3. Maintain a clean and safe work area at all times.

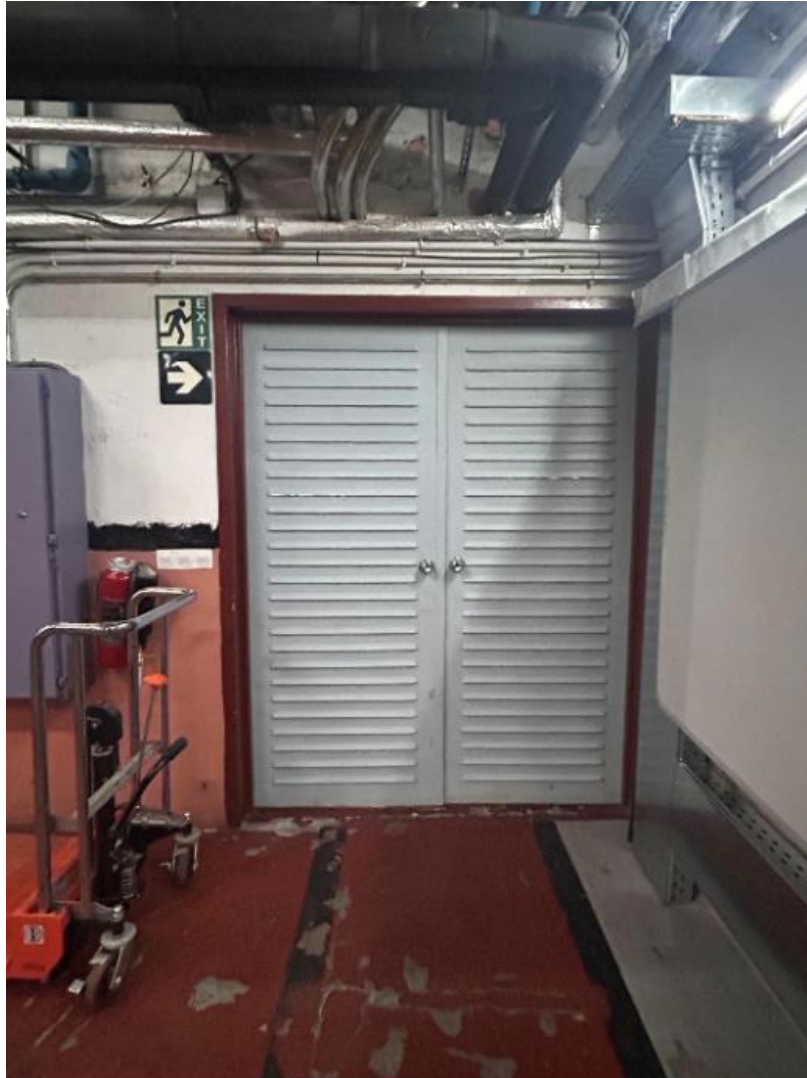
ATTACHMENT 1: DRAWINGS

A1- N1. Sub-basement floor plan. Walls to fireproof marked in pink.



ATTACHMENT 2: PICTURES

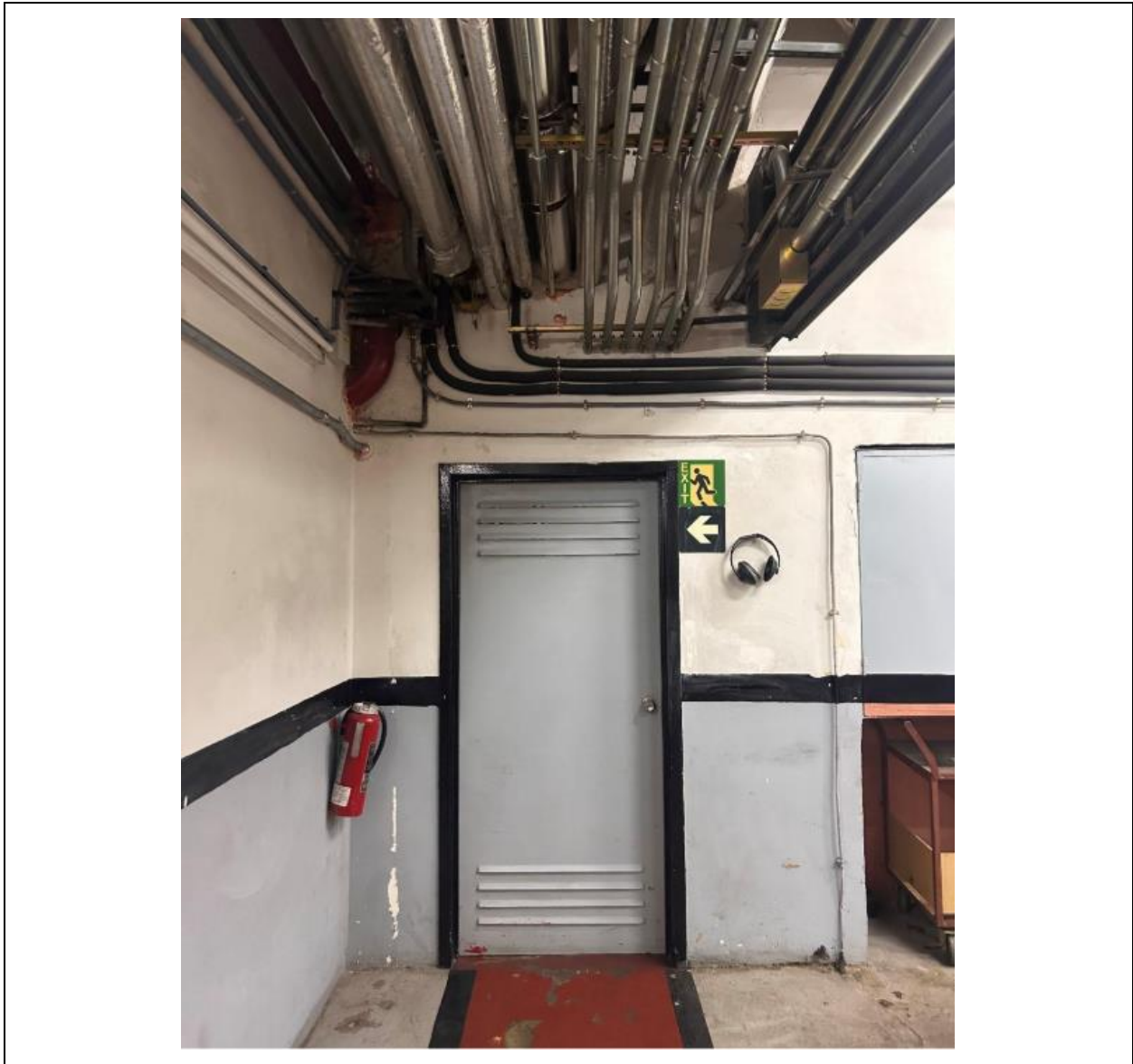
A2- N1. Door N1 – Boiler Room West



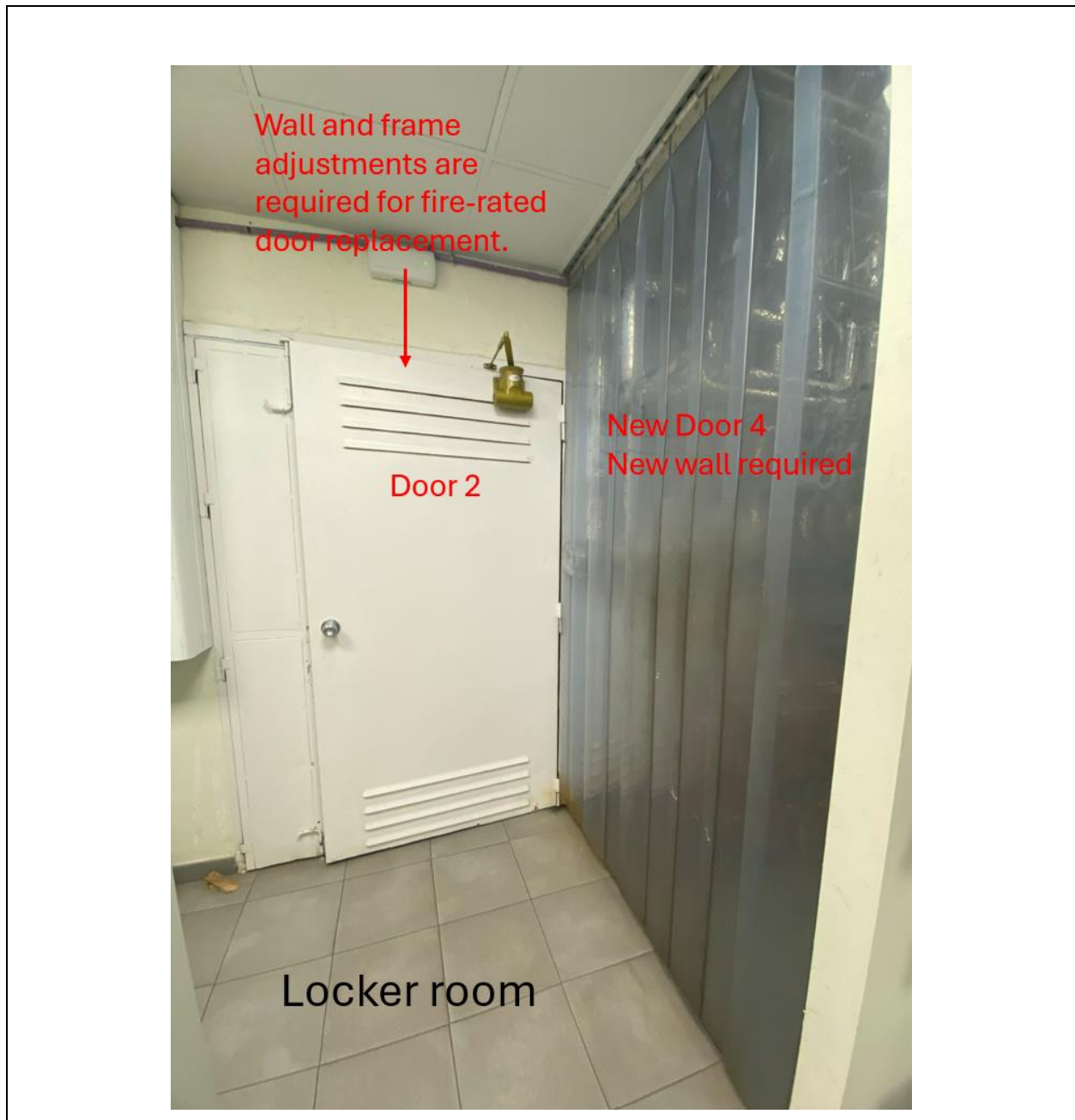
A2- N2. Door N2 – Boiler Room to locker room



A2- N3. Door N3 – Boiler Room East



A2- N4. Door N4 (New) – Location of the new fire pump door and wall. Also, in the picture Boiler room door N2 to be replaced.



A2- N5. Door N5 – High Voltage room



A2- N6. Door N6 – UPS room



A2- N7. Door 7a and 7b – Low Voltage Room

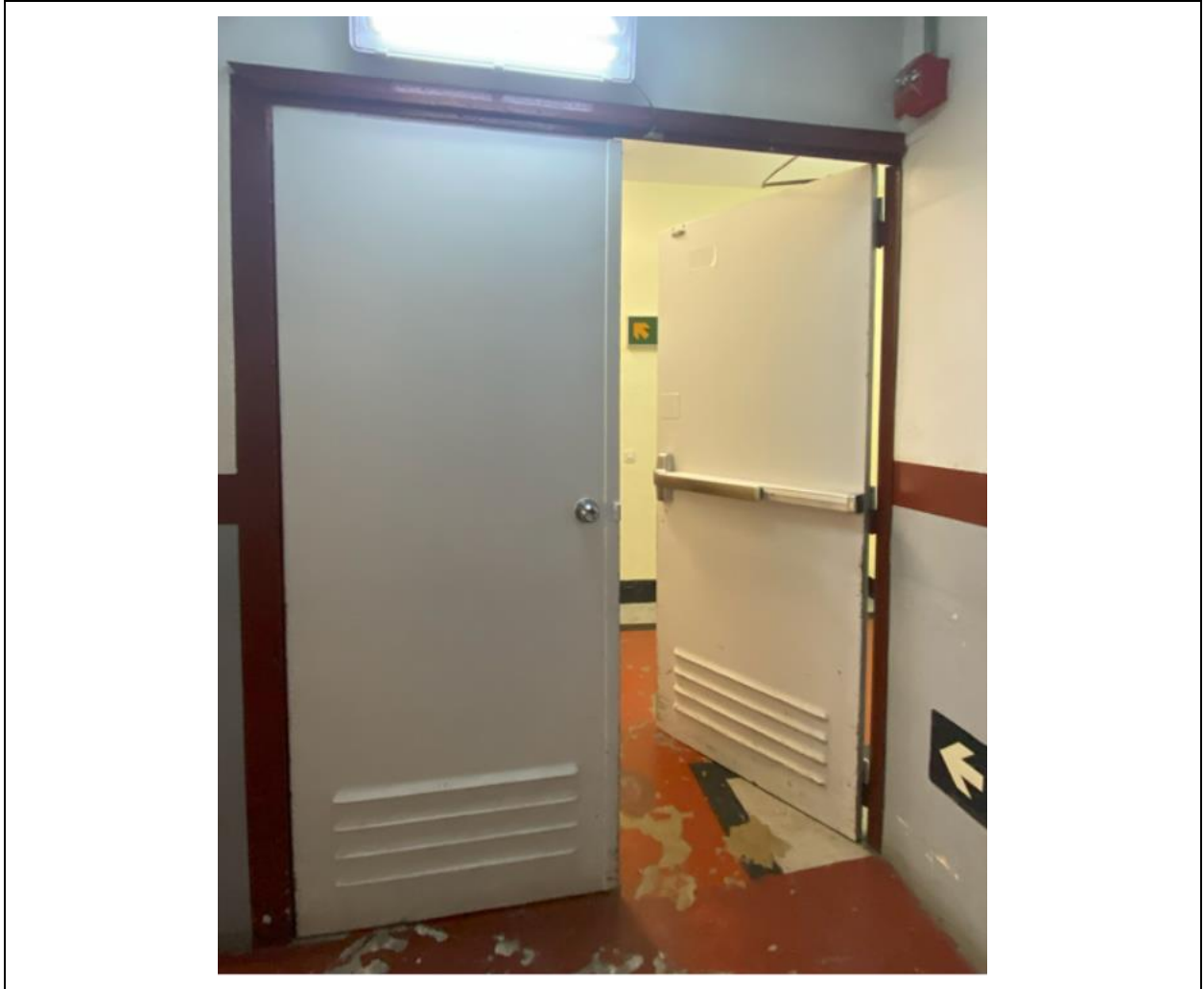
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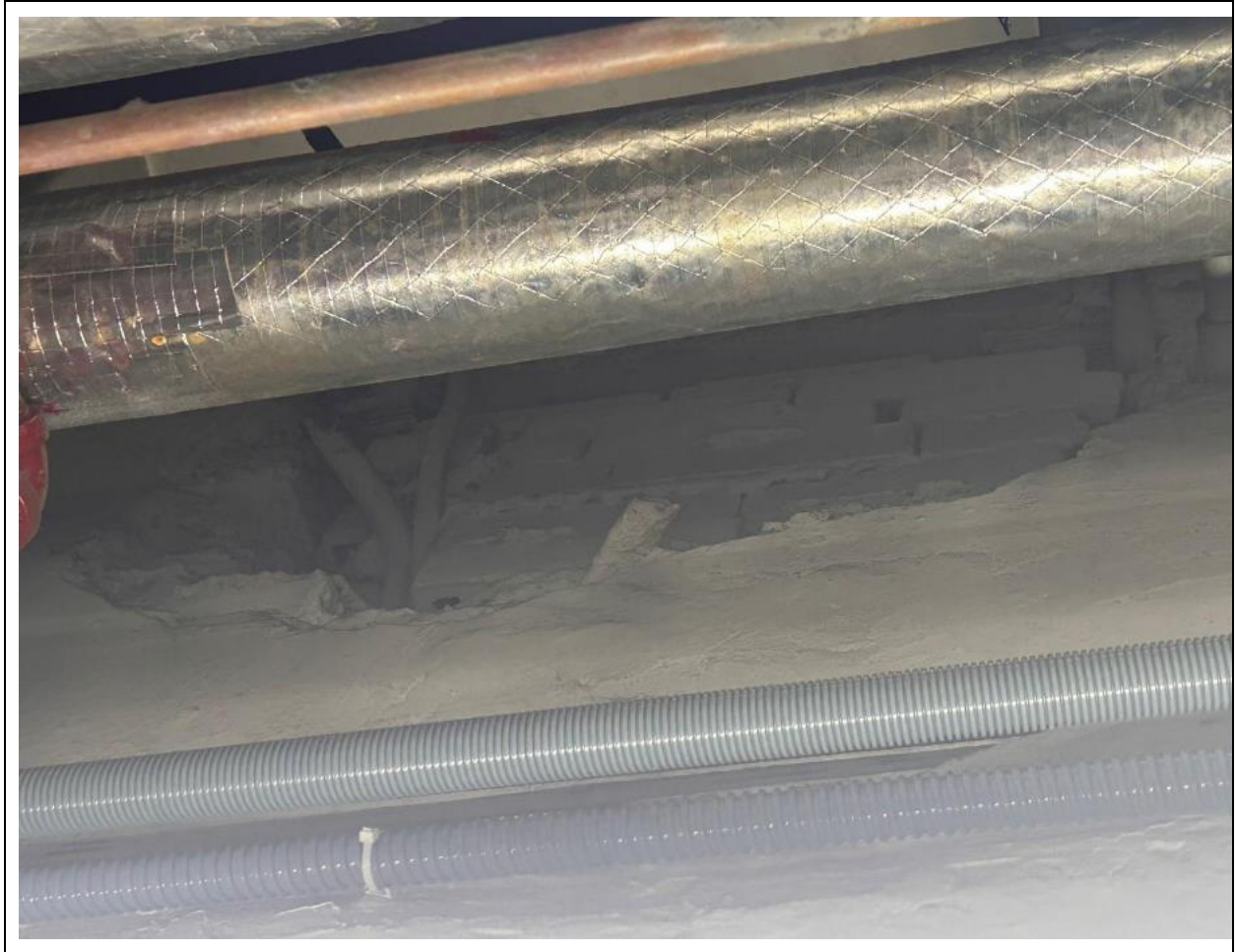
7b



A2- N8. Door N8 – Mechanical room entrance door



A2- N9. Boiler wall penetration



A2- N10. Boiler room penetrations.

